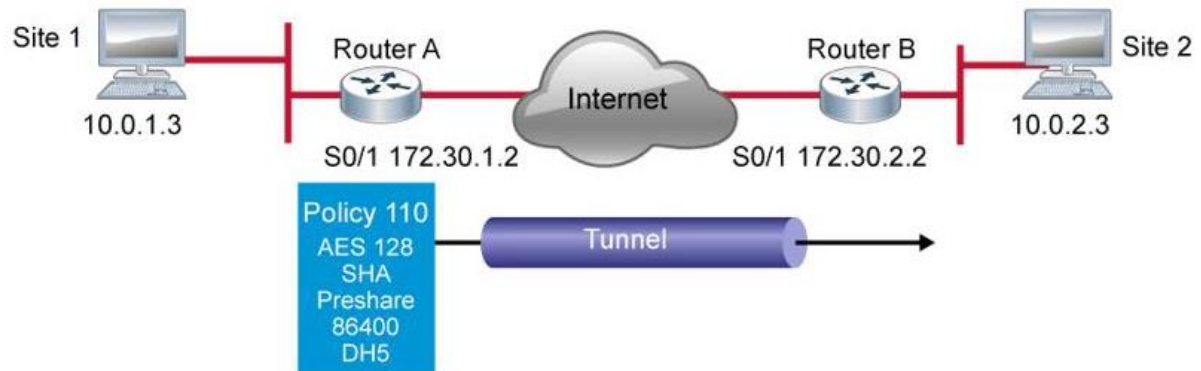


ISAKMP Policy:

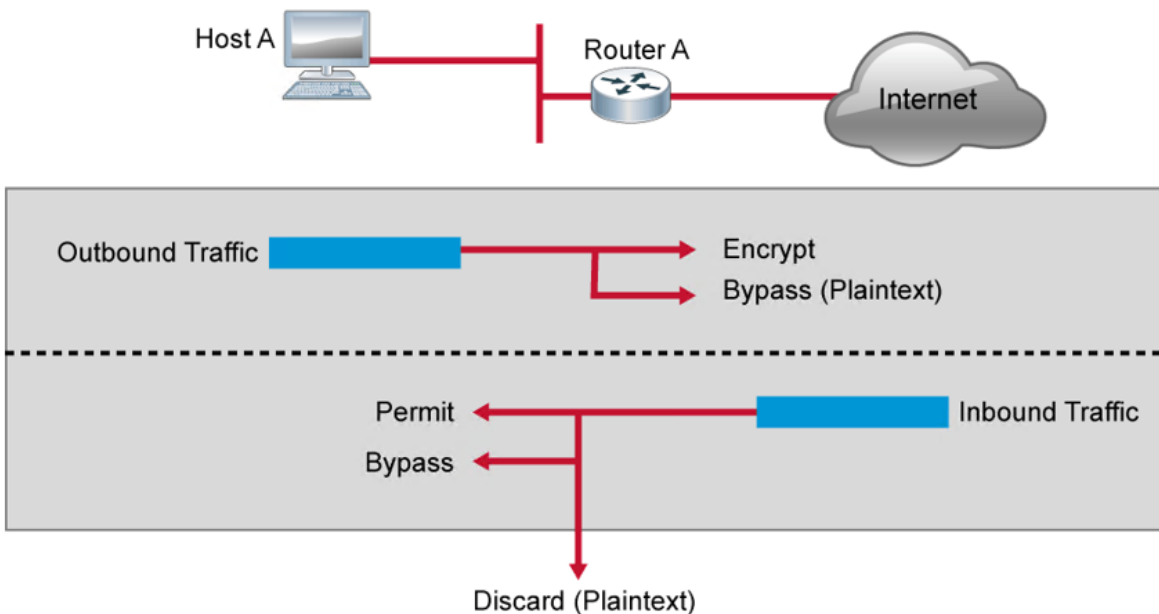
- o Defines acceptable algorithms, authentication types & lifetimes for the Phase 1 SA.



```
RouterA(config)#crypto isakmp policy 110
RouterA(config-isakmp)#authentication pre-share
RouterA(config-isakmp)#encryption aes 128
RouterA(config-isakmp)#group 5
RouterA(config-isakmp)#hash sha
RouterA(config-isakmp)#lifetime 86400
```

Encryption Domain:

- o Encryption domain specifies what traffic should be sent through the VPN tunnel.
- o The encryption domain is communicated as part of the Phase 2 exchange of VPN.
- o Any mismatch can lead to either VPN only working intermittently or not working at all.
- o Legacy VPN configuration using crypto maps defined using ACL with permit statements.
- o Specifying what be encrypted & denies being traffic that should be sent in plain text.
- o In newer IOS VPN implementations Encryption Domain is defined automatically.



Crypto Maps:

- o The primary means of linking all various components in VPN together on ASA Firewall.
- o It define peer, IPsec Proposal/Transform set, IPsec SA lifetime and encryption domain.
- o It defined globally & then linked to interface from which VPN connections established.
- o IOS VPNs still use crypto map & is still commonly done when connecting to other parties.

```
RouterA(config)#crypto map mymap 10 ipsec-isakmp
RouterA(config-crypto-map)#match address 110
RouterA(config-crypto-map)#set peer 172.30.1.2
RouterA(config-crypto-map)#set transform-set AES_SHA
RouterA(config-crypto-map)#set security-association lifetime seconds 86400
RouterA(config-crypto-map)#exit
RouterA(config)#interface serial0/1
RouterA(config-if)#crypto map mymap
```



```
RouterA(config)#crypto map mymap 10 ipsec-isakmp
RouterA(config-crypto-map)#match address 110
RouterA(config-crypto-map)#set peer 172.30.2.2
RouterA(config-crypto-map)#set transform-set AES_SHA
RouterA(config-crypto-map)#set security-association lifetime seconds 86400
RouterA(config-crypto-map)#exit
RouterA(config)#interface serial0/1
RouterA(config-if)#crypto map mymap
```



Tunnel Protection Profiles:

- o Tunnel Protection Profile link together all various components of the VPN.
- o Unlike Crypto Maps, Protection Profiles are specified on the correct tunnel interface.
- o Therefore, allow encryption to be added after the fact to other tunnel encapsulations.
- o The mode of tunneling set on the interface determines how the tunnel is implemented.
- o It's important that both peers are configured in a similar (if not exactly the same) manner.
- o Both IPsec Profile and ISAKMP Profile can be specified allowing for a greater level of
- o Flexibility in what clients should be allowed to use that specific Virtual Privat Network.
- o Tunnel Protection Profiles is used across VTI, DVTI, DMVPN & FlexVPN implementations.